Designing Rack Scale AI:

The OCP Advances its Vision of a Single "Chip to Grid" AI Production System

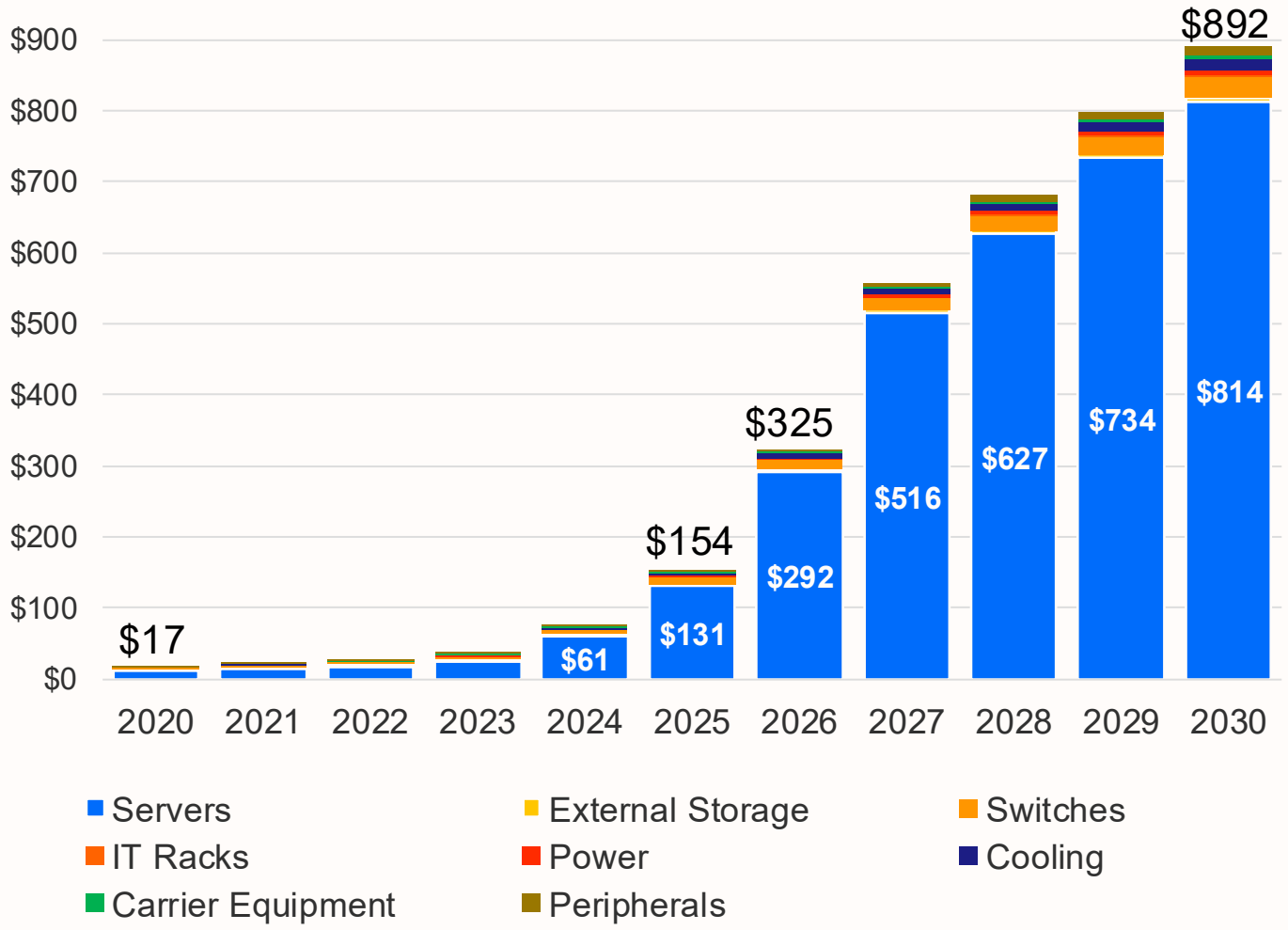
IDC Market Impact Study 2026

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Compute platform investments are driving spending across all OCP-recognized infrastructure categories



Source: IDC, 2026

- **Total spending** on OCP-recognized infrastructure and solutions will grow to US$892 billion in 2030, up from US$154 billion in 2025, reflecting a CAGR of 42%.
- Server spending (especially spending on **GPU-accelerated servers**) will account for the largest market segment and the strongest growth rate throughout the forecast.
- Servers incorporate a range of components, including accelerators, interconnects, memory devices, GPU trays, and storage interfaces.
- Spending on **power** and **cooling technologies, rack infrastructure,** and **switching equipment** will exhibit strong and stable spending growth throughout the forecast.

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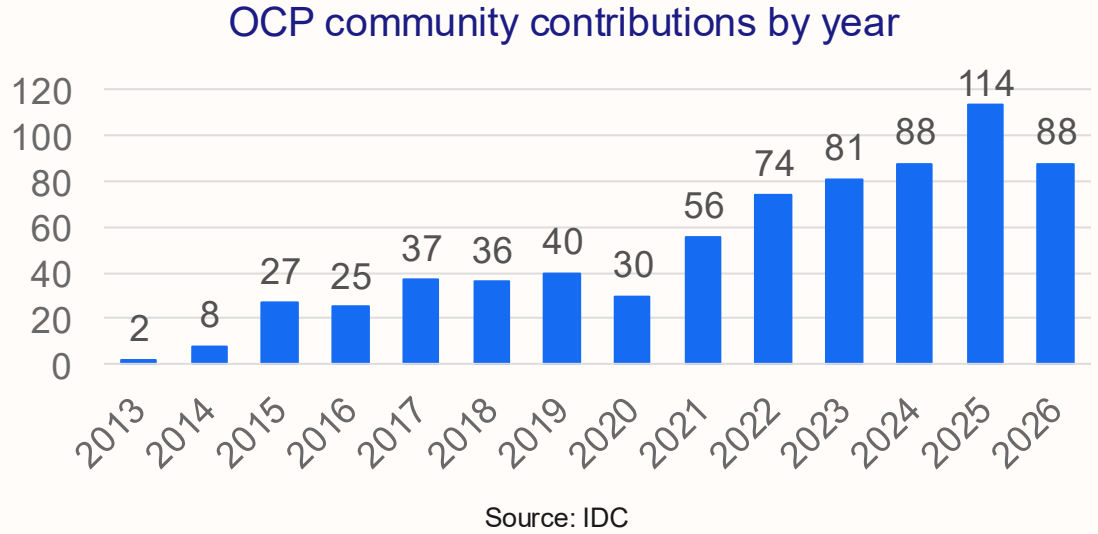
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The OCP community and vendor ecosystem continues to expand and mature, fuelling technology adoption

- In addition to **contributing** new technology specifications to the OCP Community, many members are important **end users** of OCP-recognized technologies themselves.
- Use of OCP-recognized infrastructure and technology designs goes beyond the organization's formal membership. Worldwide use of OCP-recognized IT infrastructure continues to grow thanks partly to the way hardware designs are **openly shared and available** to any organization that wants to access and implement them.
- The **ecosystem of solution providers** offering OCP-recognized infrastructure continues to grow and develop, encouraging adoption.

90%

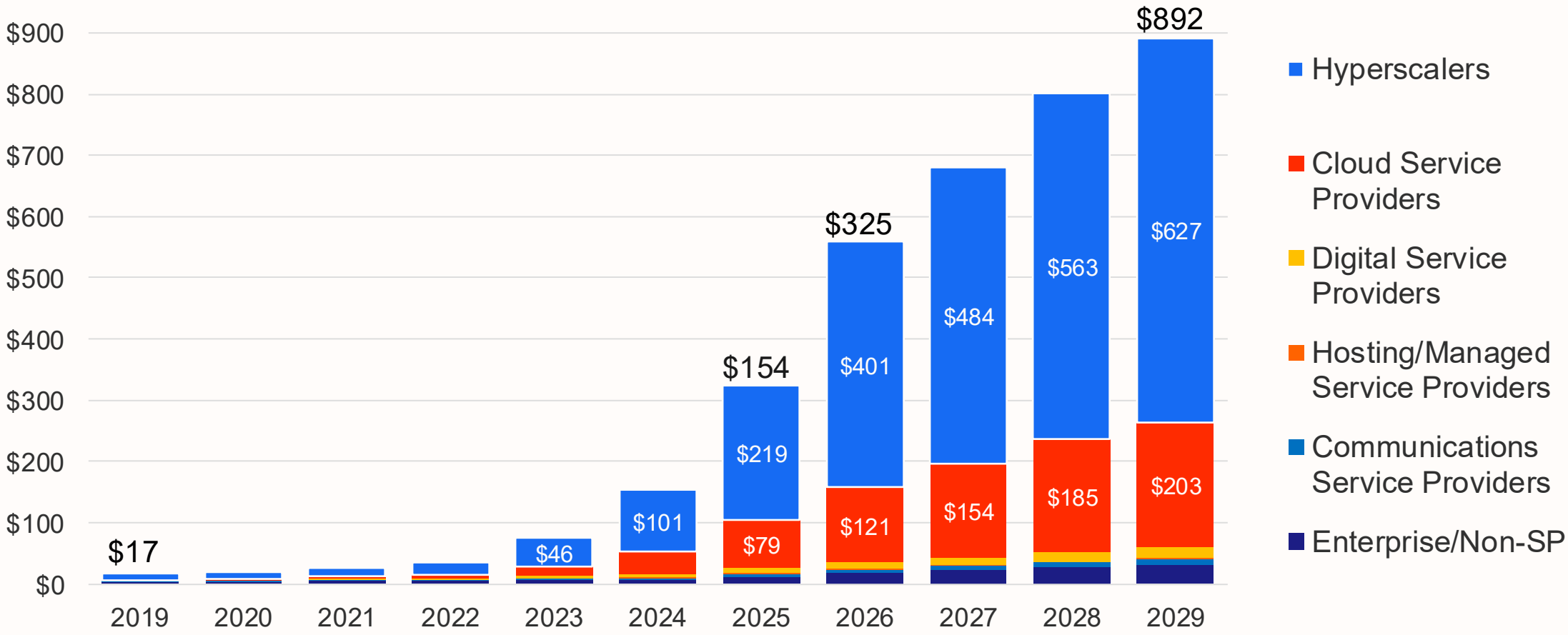
The proportion of OCP-designed servers deployed in core datacenter environments



Source: IDC

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OCP infrastructure spending by market segment, 2019-2030 (USD $B)

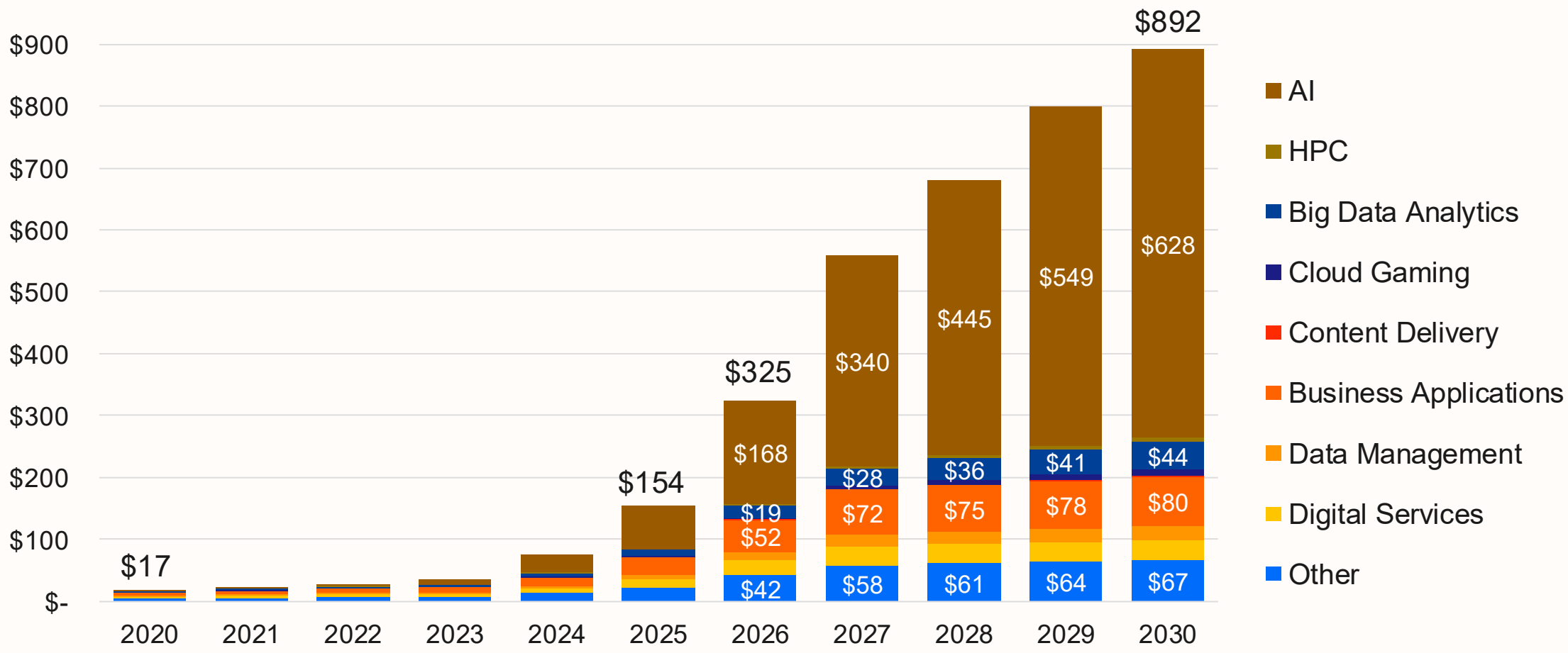


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OCP infrastructure spending by use case, 2019-2030 (USD $B)

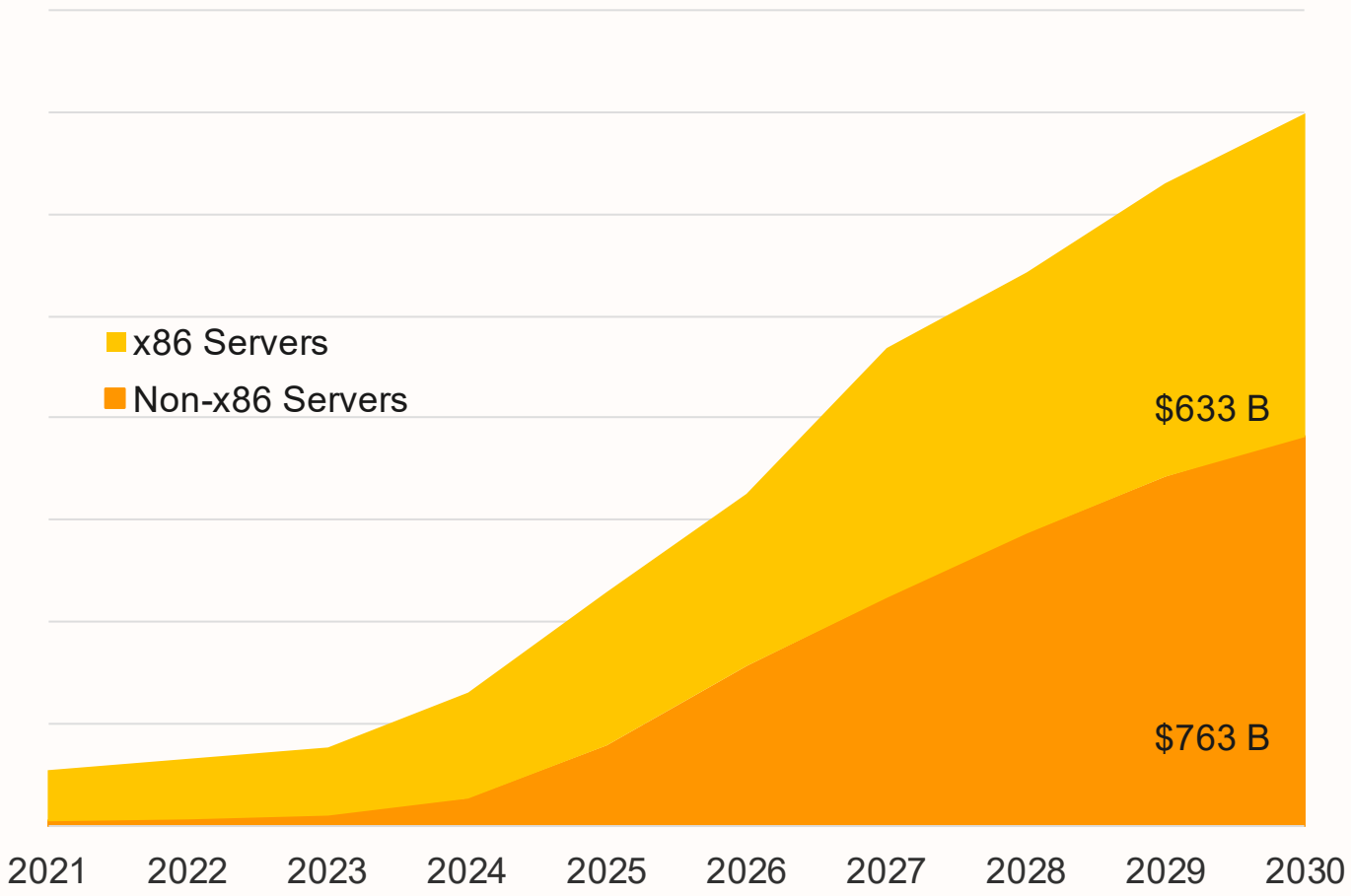


Source: IDC, 2026

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Investments in GPU-accelerated AI servers are a key contributor to AI infrastructure spending growth



Source: IDC, 2026, Enterprise Infrastructure Server Tracker, 2021–2030

- IDC's Enterprise Infrastructure Server Tracker shows that revenue from the **sale of non-x86 servers** more than doubled in 2025, compared with 2024.
- Revenue growth was driven by investments by **hyperscale** internet companies and **cloud service providers** in GPU-accelerated servers to support increasing demand from AI workloads.
- Strong growth is expected for the adoption of **GPU-accelerated servers** over the next four years, with revenue from non-x86 server sales predicted to reach $763 billion by 2030, up from $311 billion in 2026.

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Over the past year, OCP projects and work efforts have spanned compute and rack infrastructure, networking, power, and cooling technologies, and a plethora of adjacent and emerging areas.

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Rack infrastructure milestones and initiatives

Racks are transitioning from being mechanical enclosures into flexible platforms that integrate compute, cooling, power options, and telemetry

- **Accommodating wider high-density AI accelerator footprints**
  - The OCP announced the Open Rack Wide specification, a major expansion to the Open Rack ecosystem for AI, and one that aims to accommodate the extreme power, liquid-cooling, and high-density packaging needs of large-scale AI clusters (April, Barcelona).
- **Standardizing designs for AI clusters that foster a holistic "chip to grid" approach**
  - The Open Data Center for AI launched and expanded strategic guidelines (including Next Generation ML Infrastructure Design Principles) to standardize physical pods, cabling, and rack-to-rack spacing for large multi-node AI deployments (April, Barcelona).
- **Defining how ultra-dense AI and ML hardware fits into standard rack layouts**
  - Designed by Meta, the Clemente specification standardizes 1RU high-performance AI compute trays by integrating two NVIDIA GB300 Host Processor Modules and using a hybrid cooling design alongside sidecar power racks.

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Networking milestones and initiatives

Ethernet is emerging as a leading fabric for AI, which encompasses different networking scales and is based on open hardware and software

- **Adapting Ethernet for dense AI accelerator cluster interconnects**
  - The Ethernet for Scale-Up Networking (ESUN) working group (first announced in 2025) released base specifications tailored for ultra-low-latency GPU interconnect reliability (April, Barcelona).
- **Standardizing optical and photonic interconnects to overcome the limitations of copper wiring in AI datacenters**
  - The Open Silicon Photonics for AI Systems workstream aims to standardize co-packaged optics infrastructure, targeting multi-vendor specifications to replace traditional copper interconnects (August, Taipei).
- **Disaggregated switch hardware to promote interoperability and minimize lock-in**
  - Meta's Disaggregated Scheduled Fabric & Non-Scheduled Fabric architectures are designed to manage massive multi-building AI training workloads, while Minipack3N is a high-density 51Tbps switch platform that integrates advanced merchant silicon and open software stacks like SAI (Switch Abstraction Interface) and FBOSS.

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Datacenter power milestones and initiatives

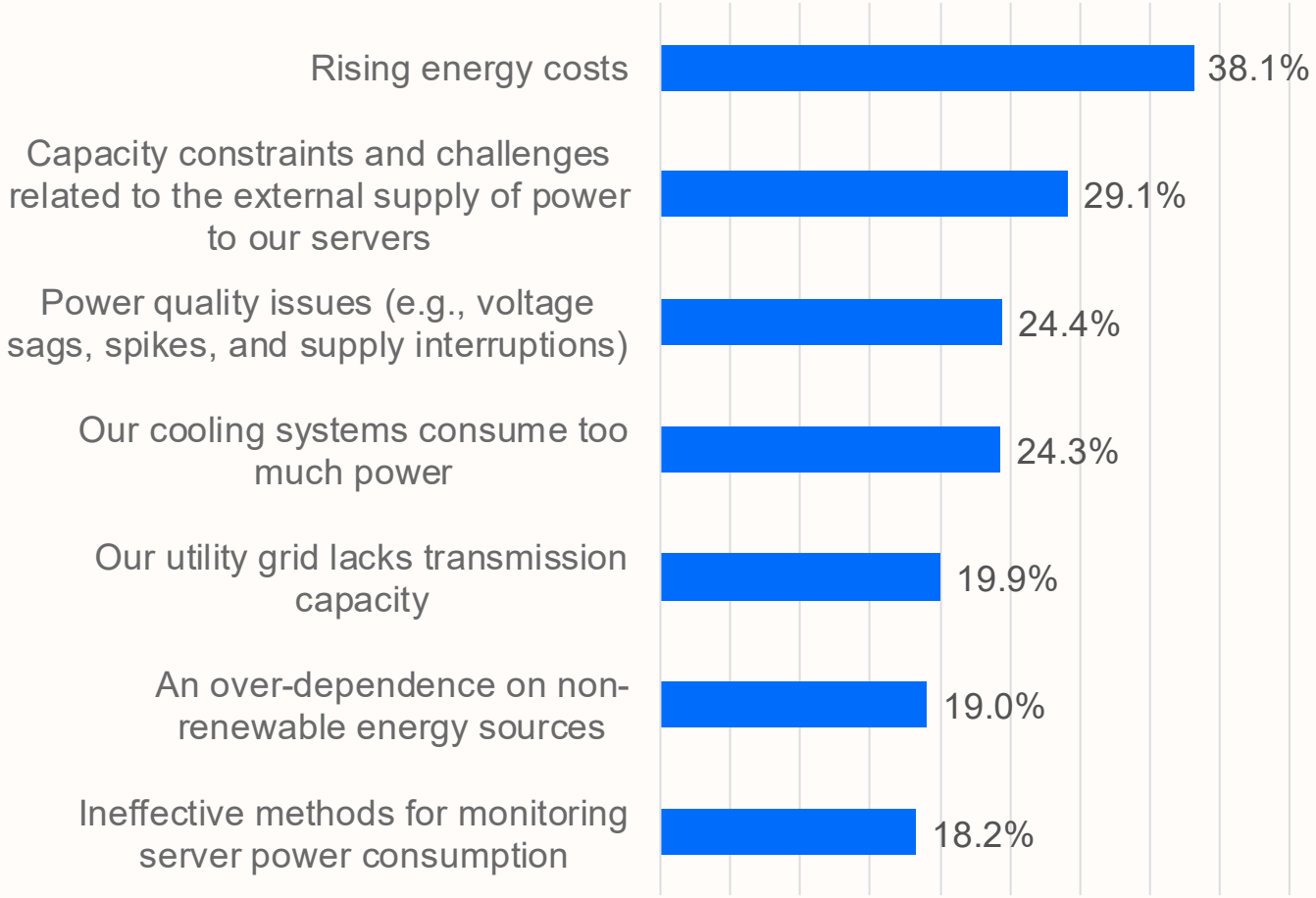
The focus is shifting from improving power delivery within the rack to redesigning the entire electrical architecture from the energy grid to the chip

- **Scaling power distribution and delivery to support ultra-dense AI clusters**
  - The OCP published low voltage datacenter power distribution framework that addresses the massive energy waste and copper bottlenecks of scaling AI racks toward 1 MW (April, Barcelona).
  - Google, Microsoft, and NVIDIA aligned on an 800V datacenter power distribution system for moving massive power loads with reduced copper loss as racks scale (August, Taipei).
- **Advancing high-voltage direct current systems for improved performance and efficiency**
  - The Low-Voltage Direct Current (LVDC) Working Group was launched in December 2025, together with Google, ABB, Siemens, and the Current/OS Foundation to standardize direct-current microgrids and power distribution safely for next-generation datacenters.
  - The Open Data Center for AI Strategic Initiative, announced in October 2025, aims to further datacenter power standards, with notable contributions including Mt Diablo (Diablo 400) power-rack sidecar for powering AI clusters (Google, Meta and Microsoft).

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Rising energy costs and supply/capacity constraints top the list of enterprise datacenter power concerns



Source: IDC, Enterprise Infrastructure Pulse Survey, 2026

- Rising **energy costs** are seen as the single biggest power-related challenge for enterprise datacenters over the next 12 months. **Capacity** and **the external supply** of power ranks as the second-biggest concern.

- Around one quarter of organizations expect **power quality issues** and the amount of power consumed by their datacenter **cooling systems** to be among their biggest challenges.

- Other power usage and supply challenges include a lack of **utility grid transmission capacity** and an overdependence on **non-renewable energy** sources.

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Datacenter cooling milestones and initiatives

Efforts to expand liquid cooling utilization aim to define the interfaces, distribution equipment, and facility architectures needed to deploy it at scale

- **Connecting servers and racks to the datacenter cooling system with CDUs**
  - The OCP launched a dedicated Coolant Distribution Unit (CDU) Project in October 2025, as part of the new Open Data Center for AI Strategic Initiative to standardize rack-level liquid cooling. Key contributions include Google's Project Deschutes CDU specification.
- **Standardizing immersion cooling as an alternative to Direct-to-Chip**
  - In 2026, the OCP Immersion Project has scaled out various parallel workstreams covering fluid lifecycle management, thermochemical reliability, sensor standardization, and component/material compatibility guidelines for dielectric fluids.
  - In May 2026, Castrol ON began certifying immersion fluids under the OCP Inspired program, making thermal management fluids directly accessible via the OCP Marketplace.
- **Enabling rack-level cooling in legacy datacenters with liquid-to-air sidecar technology**
  - In June 2026, Google unveiled plans to open-source its Brazos cooling sidecar design through OCP, providing a closed-loop 60 kW rack cooling option that allows legacy air-cooled datacenters to adopt high-density liquid cooling without costly retrofits.

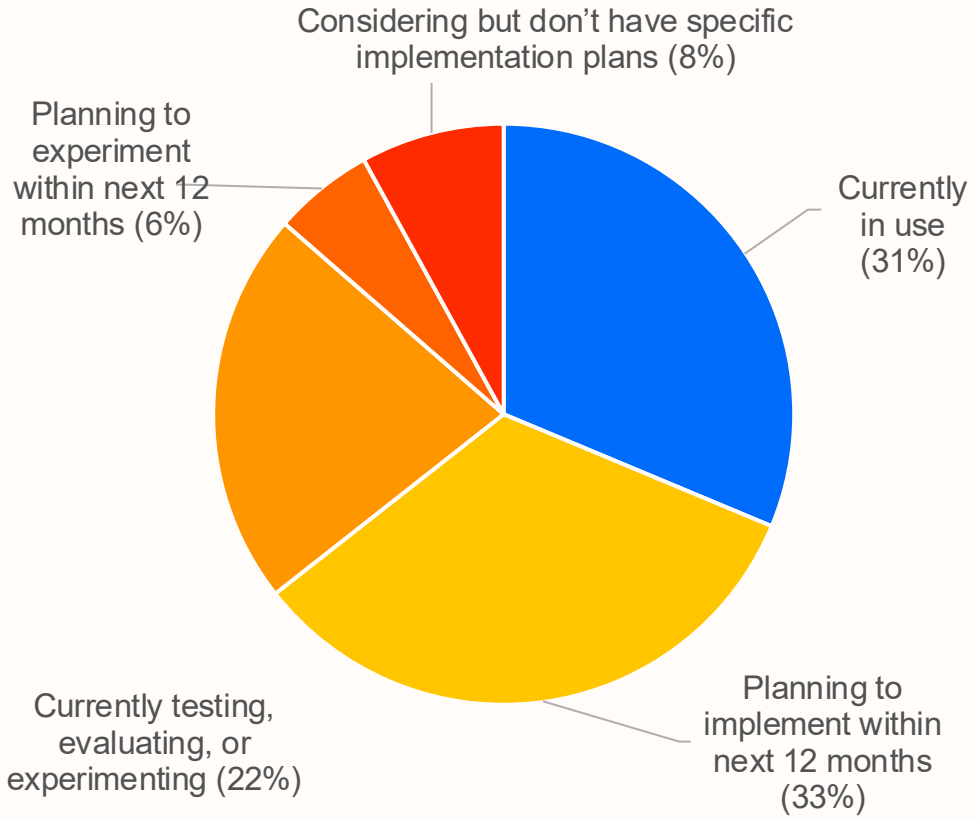
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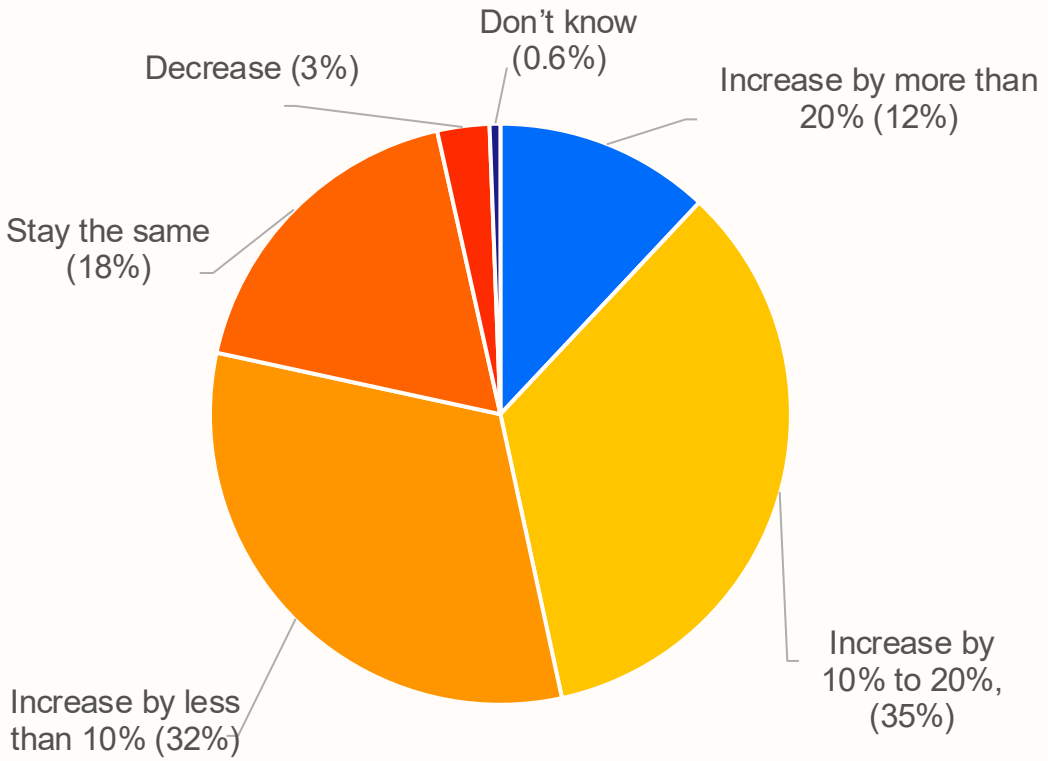


Widespread enterprise adoption of liquid cooling is already underway with clear plans to expand usage

Q. Which of the following best describes your use or planned use of liquid cooling?



Q. In the next 12 months, do you plan to increase, decrease, or maintain its usage of liquid cooling?



Source: IDC, Enterprise Infrastructure Pulse Survey, 2026

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Emerging workstreams and initiatives

The OCP's vision is expanding from open hyperscale infrastructure into a broader program for the next generation of AI infrastructure

- **Preparing for the deployment of Quantum Processing Units (QPUs) within commercial, production-scale datacenters**
  - Announced in April 2026, the OCP's Quantum Integration Initiative aims to establish a standardized engineering blueprint that will enable organizations to harness the benefits of QPUs within their datacenters.
  - Addresses key infrastructure challenges including weight and floor loading requirements, thermal and cooling demands, and the need for thermal isolation.
- **Strengthening security evaluations for datacenter hardware and firmware**
  - The OCP announced the growing ecosystem adoption of the Security Appraisal Framework for Independent Firmware Evaluation (S.A.F.E.), which standardizes and simplifies the operation of hardware and firmware security (August, Taipei).
  - S.A.F.E. aims to replace fragmented, repetitive evaluations with a single, unified device-specific checklist managed by certified third-party Security Review Providers. A supply chain focus is intended to help cloud providers and enterprises verify the security posture of complex silicon and firmware components before deployment.

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Summary and takeaways

- **Upstream focus on technology and supply chain standardization**
  - The focus has moved upstream in response to the computational requirements of AI. This extends from the silicon-level (comprising individual microchip components) to the facility-utility level, with an emphasis on the supply and distribution of power to and within the datacenter.

- **Integrating reference architectures and validating single system AI**
  - To date, there has been a strong focus on defining the open building blocks, but going forward, the emphasis is likely to be more about integrated reference architectures and validating AI infrastructure clusters that bring together different specifications within a single system.

- **Deepening and demonstrating interoperability**
  - Increasingly, success and progress will be measured by the extent of interoperability, with the OCP strengthening its value as a neutral environment where competing suppliers can demonstrate that their products work together.

- **Broadening the open ecosystem's deployment environments**
  - The OCP's AI Computing Continuum will continue to advance specifications that broaden the open ecosystem beyond the traditional hyperscalers. This will require specifications that accommodate a wider range of rack sizes, cooling environments, networking topographies etc.

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Key takeaways

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